

Project Overview

With the help of SQL for querying and PowerBI for visualization, I wanted to understand my personal listening history a bit more. I have been using Spotify for a decade now, and wanted to see how much I have changed since then. Tracking all types of changes from my middle school years up until my current graduate lifestyle.

Dataset Description

The data was obtained via my Privacy Settings page in my personal Spotify account. This history goes back to 2015 when I first opened my account, until the beginning of 2026, containing over 100k+ rows of data.

Total Streams (Num) - Shows the count of all songs played

Total Listened Time (Num) - Shows the sum of all minutes played

Song (Text) - Displays each song name that has been recorded

Listening Time (Num) - Displays the total sum of minutes a specific song has been played

Artist (Text) - Displays each artist's name that has been recorded

Album (Text) - Displays each album's name that has been recorded

Skipped (Boolean) - Records when a song was played for less than 30 seconds

Unique Songs (Num) - Shows the count of all distinct songs played

Hours (Time) - Displays a 24:00 hour clock

Years (Time) - Displays each year that data has been recorded

Data Cleaning Process

```
select skipped, shuffle, reason_start, reason_end,
       master_metadata_album_album_name as album,
       master_metadata_album_artist_name as artist,
       master_metadata_track_name as song,
       ms_played / 60000.0 AS minutes_played, sh.offline,
       date(ts) as date,
       time(ts) as time
from streaming_history sh
where master_metadata_track_name != ''
```

First step was to convert JSON files to CSV one by one, to be able to use SQL to clean the data. It had come with several extra columns, such as information about podcasts and audiobooks, which were unnecessary for what I wanted to analyze in this project. The minutes played (of each stream) had also come in milliseconds, which meant I had to convert it to minutes and hours for better readability. Since Spotify's Personal Streaming Data comes with Spotify's own column names (they are very long), I changed them to a visual-friendly name for easier use. Lastly, made sure not to include any null values.

Business Questions

"Which artists do I listen to the most?"

"Which were my most streamed songs throughout this decade?"

"Which album do I replay the most?"

"What time of day did I listen to music the most?"

“How many new songs were streamed?”

“How many tracks were skipped throughout the year?”

SQL Queries

```
-- Which songs have the highest total listening time?
select master_metadata_track_name as song,
       Round((sum(ms_played) / 60000.0),1) as minutes_played,
       Round((sum(ms_played) / 360000.0),1) as hours_played
from streaming_history sh
where master_metadata_track_name != ''
group by 1, master_metadata_album_artist_name
--this is because there are songs with the same name from different arists, so songs with different arists are counted in different places
order by 2 desc
limit 10
```

Changed the milliseconds given by Spotify to minutes and hours. Then used the Round datatype to see a clearer value for each song in the top 10. Had to group by artist, since there are several song names with the same name, and I wanted a clear view of each specific song's streamed time. Similar logic was used to find top albums, artists, top streamed hour, and total streamed time.

```
--Unique Tracks per Month/Year:
select STRFTIME('%Y', ts) AS year, STRFTIME('%m', ts) AS month,
       count(distinct sh.master_metadata_track_name || master_metadata_album_artist_name) as unique_track,
       count(*) as total_streams
from streaming_history sh
where sh.master_metadata_track_name != ''
group by 1,2
order by 1,2
```

Similar as the above query, just added distinct clause to count only the tracks played for the first time. The time function from SQLite was also used to find the year, and month (this was also used for top streamed hour and total streamed time). Also counted each track played in a separate column. The query was grouped and ordered by year, then month.

```
⊖ -- Which day of the week do I listen the most?
Select case strftime('%w', ts)
       when '0' then 'Sunday'
       when '1' then 'Monday'
       when '2' then 'Tuesday'
       when '3' then 'Wednesday'
       when '4' then 'Thursday'
       when '5' then 'Friday'
       else 'Saturday' end as weekday,
       ROUND(SUM(ms_played)/60000.0,1) AS minutes_played,
       count(*) as total_streams
from streaming_history sh
where sh.master_metadata_track_name != ''
group by 1
order by 2 desc
```

In this query, the logic had to change a bit, since I needed the output to display each day of the week, not just SQLite designated number per day. I used the case statement to change each weekday, and made the last day ('Saturday', in this scenario) the else clause. Aside from this detail, the rest of the query has the same logic as the last few. The sum of all

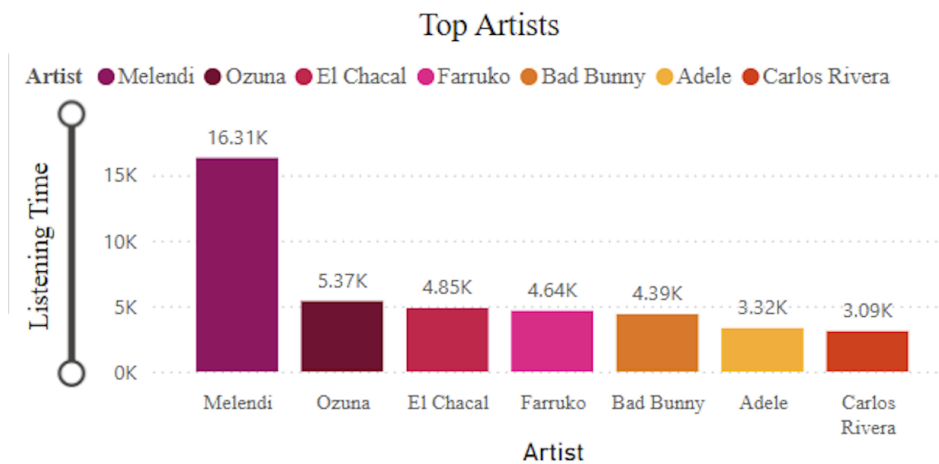
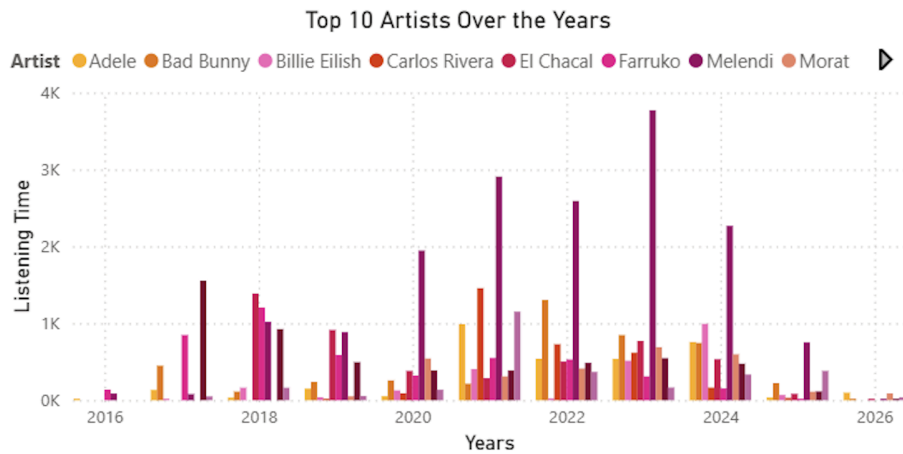
milliseconds converted to minutes, which was then rounded, and the count of all tracks played per day. Grouped by the week day, in the order of more minutes played.

```
--Time Analysis
-- How has my listening activity changed over time?
```

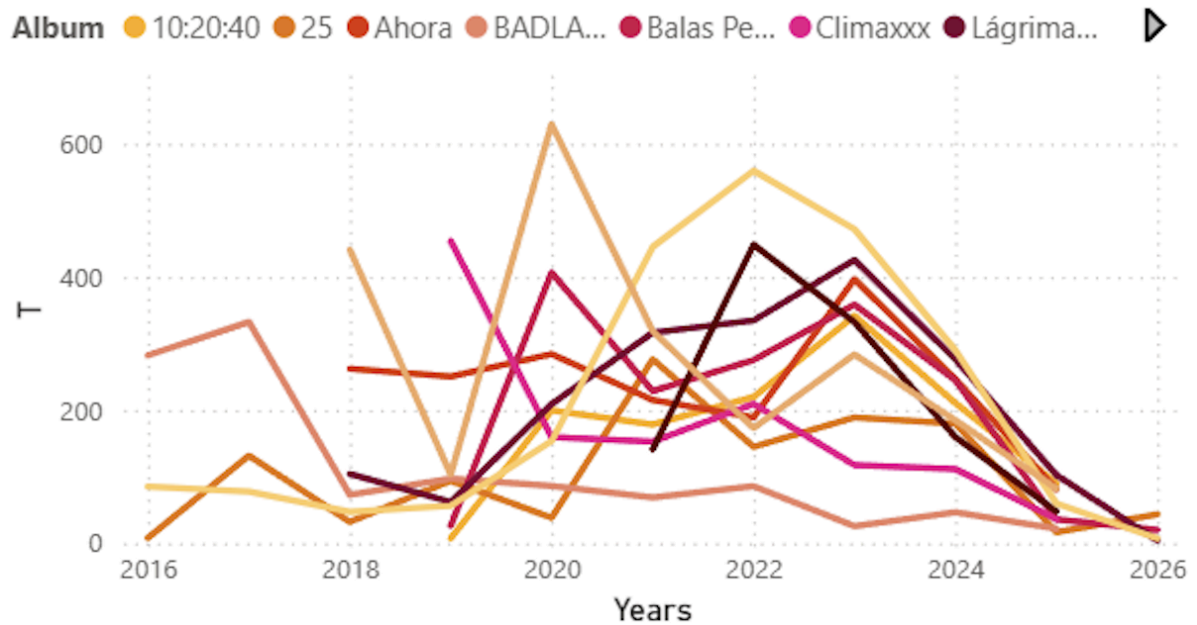
```
⊖ --Top Artist per Month of Each Year:
WITH monthly_artist AS (
    SELECT
        STRFTIME('%Y', ts) AS year,
        STRFTIME('%m', ts) AS month,
        master_metadata_album_artist_name AS artist_name,
        SUM(ms_played) AS total_played
    FROM streaming_history
    WHERE master_metadata_track_name != ''
    GROUP BY year, month, artist_name
),
ranked AS (
    SELECT *,
        RANK() OVER (
            PARTITION BY year, month
            ORDER BY total_played DESC
        ) AS rnk
    FROM monthly_artist
)
SELECT
    artist_name,
    month,
    year,
    COUNT(*) OVER (PARTITION BY artist_name) AS times_top_of_month
FROM ranked
WHERE rnk = 1
ORDER BY year, month;
```

This was the last query I wrote for this analysis. I wanted this to showcase my top artist per each month of each year; while simultaneously including how many times this artist has been top one for the month in a different column. Since what I wanted meant I needed to group by the totals but also keep the details of the rows, I decided to use CTEs and window functions. First `monthly_artist` builds the data into a table, which was done in the above queries. Then the `ranked` table adds the artists rank on a column next to the data gathered in `monthly_artist`. The window function here restarts the ranking for each month, and orders the data in descending order, to find the top artist. Lastly, we filter only by `rank = 1`, then count how many times an artist has come up as top of the month. This query was ordered by the year and month, to accurately see my music taste progress.

PowerBI Dashboard

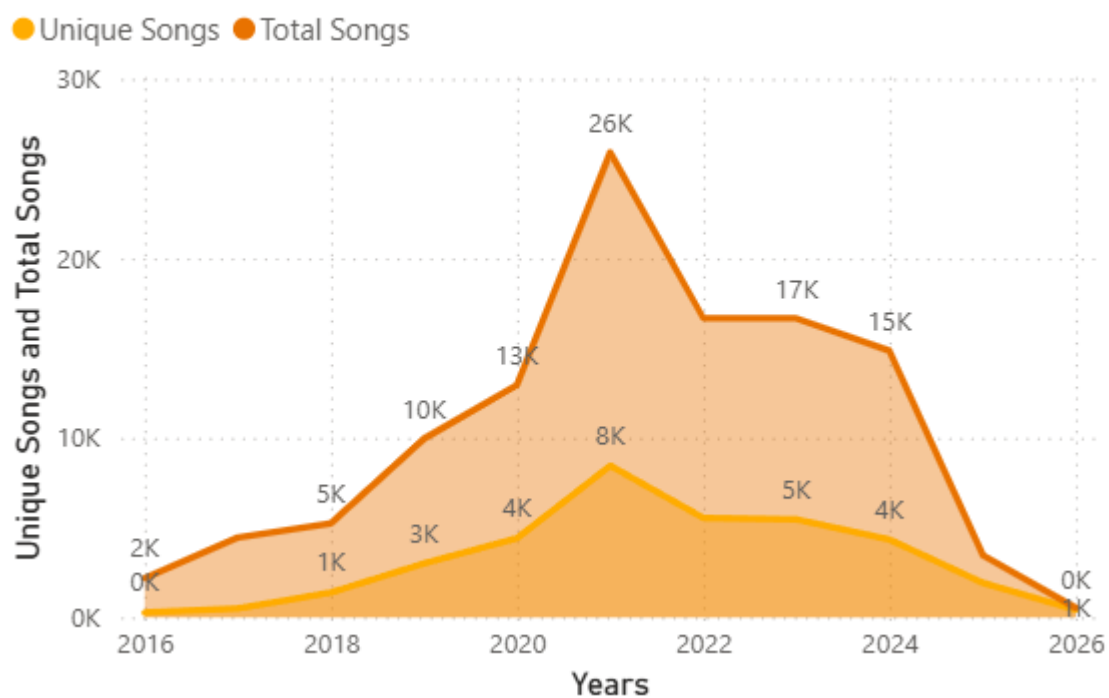


A Clustered Column Chart (1), which was selected because it best demonstrates the evolution of my top 10 artists of all time, over this past decade. I wanted to compare the streamed time between artists and their streamed rate over the years. I set a TopN filter on the artists column, to only analyze my top 10. Then a Stacked Column Chart (2), to just see the total listened time per artist from the top 10, in descending order.



Same thing as above, I wanted to see the peak years where I streamed a specific album, and the trend each of these albums had over this decade. For that, I chose a Line Chart with a TopN filter set to my top 10 most streamed albums. Here I can easily see when an album had a constant stream rate, or when I streamed it for a long time, or not as much.

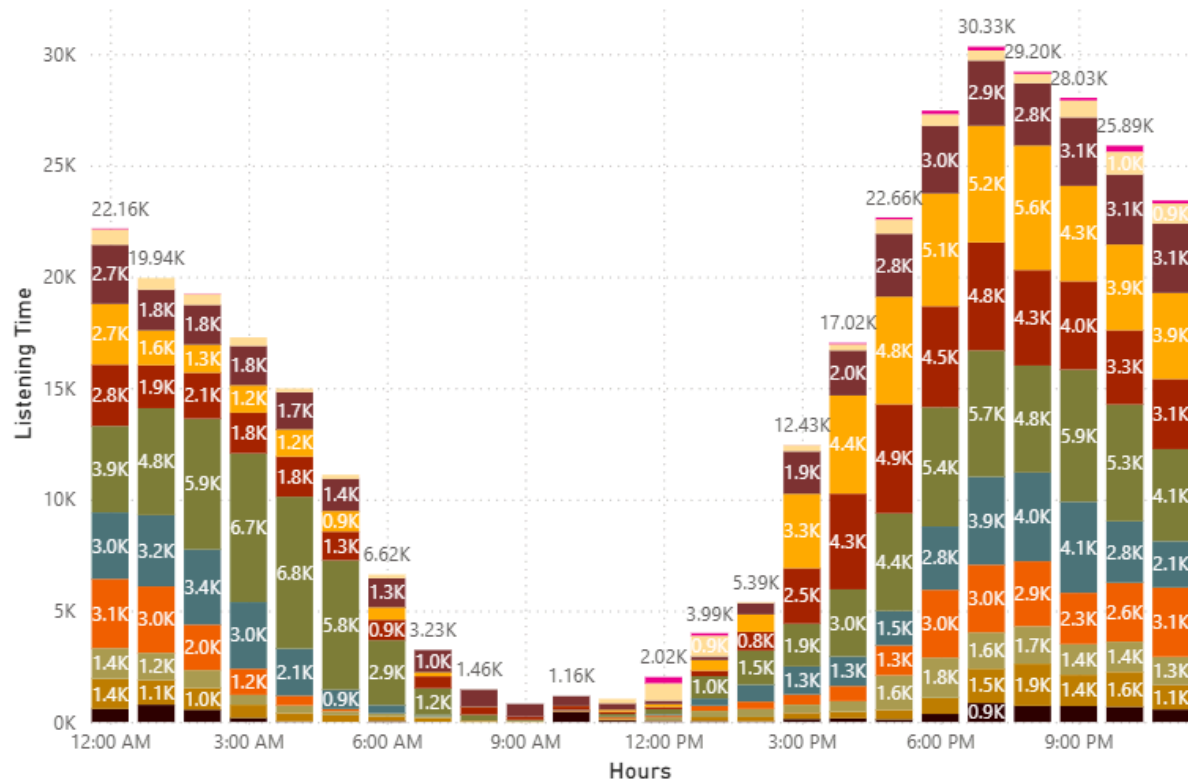
Unique Songs per Year



For this one, I wanted to see a comparison of how many new songs I was streaming compared to songs I kept repeating. This also let me see the top amount of tracks, per each year I was analyzing. For this reason using an Area Chart was the best way to compare these two trends.

Time of Day Most Streamed

Years ● 2016 ● 2017 ● 2018 ● 2019 ● 2020 ● 2021 ● 2022 ● 2023 ● 2024 ● 2025 ● 2026



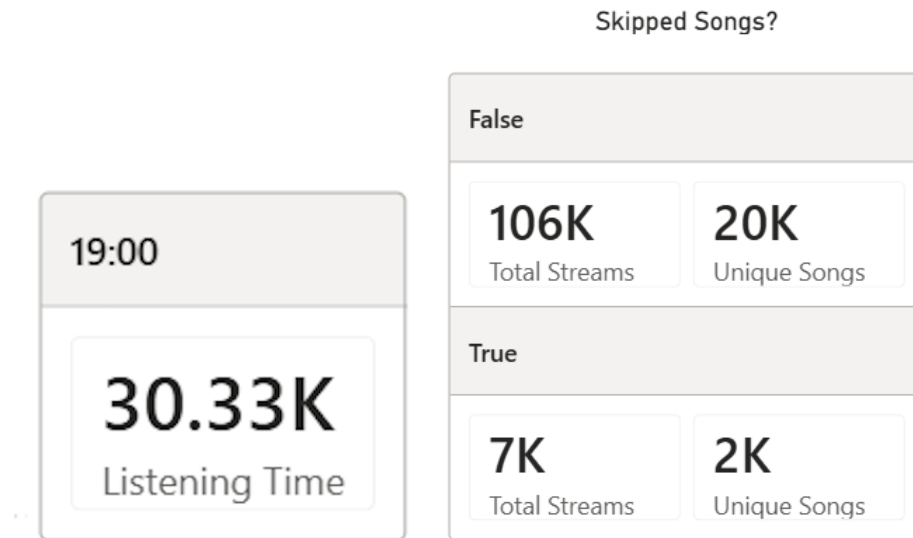
I used a Stacked Column Chart to be able to see what time of day was most streamed, per year. From the other visuals, this one displayed it the best, as there are 24 hours in a day and 10 years to compare. Had to use a different mix of colors for this one, as the color theme was too monotone to accurately compare each year's data.

113K

Total Streams

346.87K

Total Listened Time



Card Visuals were used to showcase the top value for what total values I considered most important in my analysis. This clearly displays how much of my time I have used to stream music and how many total tracks I've played. I also used it to display how many songs were skipped based on boolean data Spotify provided. It perfectly displayed how many tracks were skipped or not.

Key Insights

- My most streamed artist was top one for five consecutive years, having 3 times the total streamed time than the second most listened artist.
 - 50% of my top 10 most listened songs are from my top one artist.
 - 53% of my top 13 most listened albums are also from my top one artist.
- 10% of new tracks I listened to, did not stay on for longer than 30 seconds.
- About 20% of my total streams were new tracks; leaving the other 80% of my total tracks on repeat.
- 2021 was by far my most streamed year, since I graduated high school and had more available time.
 - 2025 had a dramatic drop of 20% of streamed time just from 2024 as I got accepted into a new job that has a no-phone-policy.
- 8-10 am were the least streamed hours of the day per the entire decade of data.
 - 7-10 pm were the most streamed hours of the day per the entire decade of data.
- In 2021, 2-4 am were the most streamed hours of the year as I graduated high school and had more autonomy over my time.
 - In 2023, 6-8 pm were the most streamed hours of the year as I played music for my previous job during rush hour on most days.